



# AI ENGINEER CAREER ROADMAP

Personalized 6-Month Strategy • Working Professional Track • Chain-of-Thought Analysis

TIMELINE  
6 Months

LEVEL  
Intermediate

TRACK  
AI Engineer

## CURRENT POSITION

Role: Working Professional  
Skills: Python · ML · Data Analysis  
Database: SQL · Cloud (AWS/GCP/Azure)  
Foundational: Mathematics & Statistics

## TARGET GOAL

Goal: Become an AI Engineer  
Focus: LLMs · RAG · MLOps · Agents  
Salary Target: ₹25–40 LPA (India) / \$130K+ (Global)  
Timeline: 6 Months Aggressive Track

## SKILL GAP ANALYSIS

### YOU ALREADY HAVE

Python · ML · Data Analysis  
SQL · Cloud · Math & Stats

### GAPS TO FILL FAST

LLM APIs (OpenAI / Gemini)  
LangChain · LlamaIndex · RAG

### ADVANCED TARGET

MLOps: Docker · Kubeflow  
AI Agents · Vector DBs

### DIFFERENTIATORS

System Design for AI Products  
Open-source Contribution

## RECOMMENDED LEARNING PLAN

Month 1–2

### LLM Foundations

OpenAI API · Prompt Eng · HuggingFace · Fine-tuning basics

Month 2–3

### RAG & Frameworks

LangChain · LlamaIndex · Vector DBs (Pinecone/Weaviate)

Month 3–4

### MLOps & Deployment

Docker · FastAPI · CI/CD · AWS SageMaker · Monitoring

Month 4–5

### AI Agents & Advanced

AutoGen · CrewAI · Multi-agent systems · Evals

Month 5–6

### Portfolio & Job Hunt

GitHub polish · Resume · Mock interviews · Applications

## SUGGESTED PROJECTS

01

### RAG Chatbot

PDF Q&A bot with LangChain + Pinecone + Streamlit UI

02

### AI SQL Agent

NLP-to-SQL agent using your existing SQL expertise

03

### Fine-tuned LLM

Domain-specific model on HuggingFace (healthcare/finance)

04

### MLOps Pipeline

End-to-end CI/CD with Docker, FastAPI & GitHub Actions

05

### Multi-Agent System

AutoGen agents that research, plan & execute tasks

## NETWORKING STRATEGY

LinkedIn

Post AI project builds weekly — show, don't just tell

GitHub

Maintain active streak; star & contribute to AI repos

Twitter/X

Follow: Andrej Karpathy, Harrison Chase, Yann LeCun

Communities

Join: Latent Space, AI Engineer Discord, Kaggle forums

Cold DMs

Reach 3 AI engineers/week; ask for 15-min coffee chats

## MONTHLY MILESTONES

M1

Complete LLM course + build first API-powered chatbot

M2

Deploy RAG app publicly + 500 LinkedIn followers gained

M3

Fine-tuned model on HuggingFace + 2 GitHub projects live

M4

MLOps pipeline running + first technical blog published

M5

3 AI agent projects done + resume + 10 applications sent

M6

Land AI Engineer offer! Negotiate. Celebrate. Repeat.

## IMMEDIATE NEXT ACTIONS (Start TODAY)

1

Enroll in DeepLearning.AI 'ChatGPT Prompt Engineering' free course (2 hrs)

2

Set up OpenAI API key & build your first LLM script tonight

3

Update LinkedIn headline to: 'ML Engineer → AI Engineer | LLMs · Python · Cloud'

4

Star LangChain & LlamaIndex repos on GitHub — read their quickstart guides